Precalculus Review for Calculus

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Functions and composition

Let $f(x) = \sqrt{x - 1}$ and $g(x) = x^2 + 3$. Find:

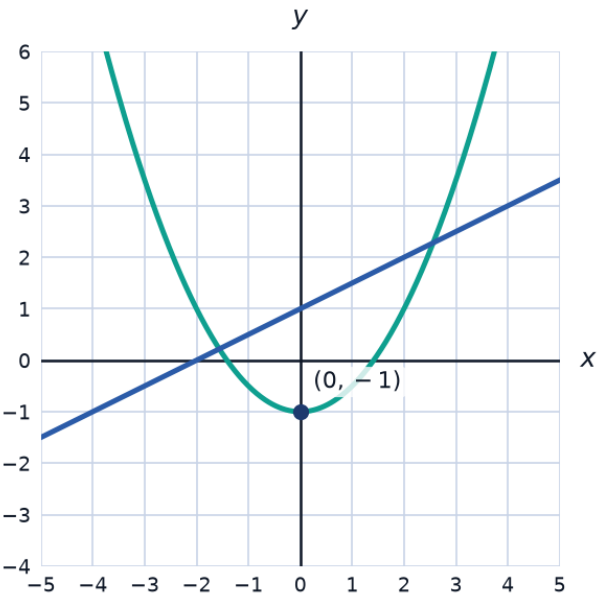
a $(f \circ g)(x)$

b $(g \circ f)(x)$

c the domain of $f \circ g$

2 [Foundational] • Target: Functions and composition

Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.



3 [Foundational] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.

4 [Foundational] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

Solve $2e^{3x} = 14$ exactly and to three decimal places.

5 [Foundational] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

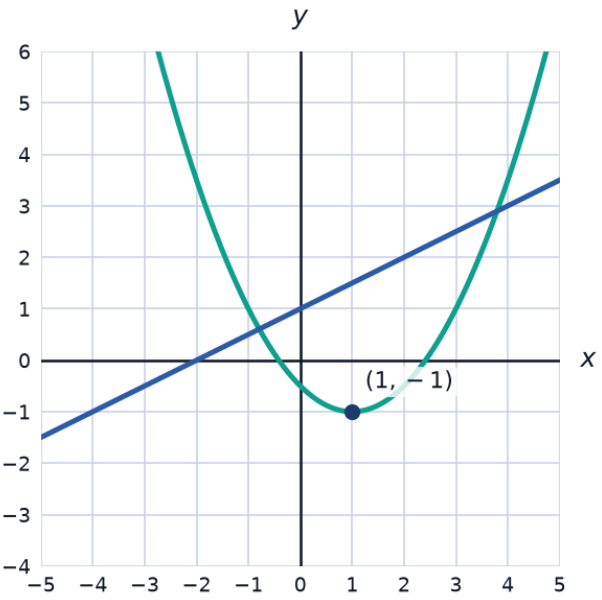
State the exact value of each expression, needed frequently when differentiating trig functions:

a $\sin\frac{\pi}{4}$ **b** $\cos\frac{\pi}{3}$ **c** $\tan\frac{\pi}{6}$

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6 [Foundational] • Target: Factoring and rational expressions

Factor each expression fully:



a $x^2 - 5x + 6$ **b** $4x^2 - 9$ **c** $x^3 - 8$

7 [Foundational] • Target: Factoring and rational expressions

Simplify $\frac{x^2 - 4}{x^2 - x - 6}$, stating the value(s) of x excluded from the domain.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inverse functions and logarithms

Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.

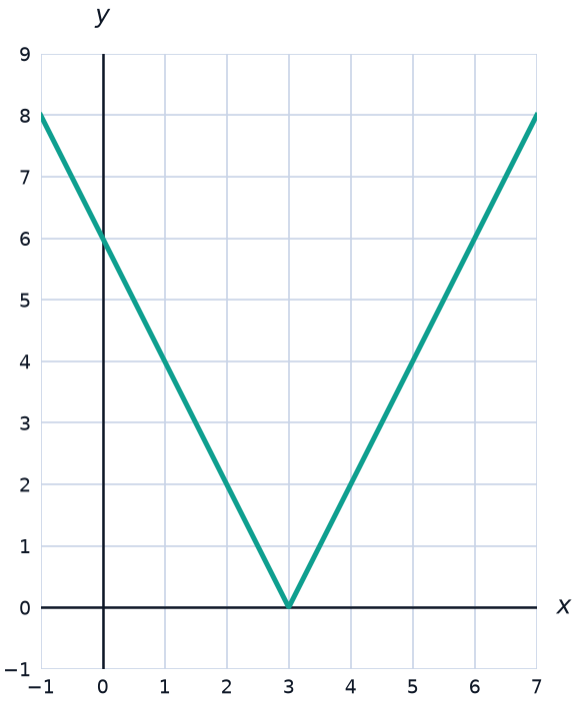
9 [Intermediate] • Target: Inverse functions and logarithms

Solve $\ln(x - 1) = 2$ exactly and to three decimal places.

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10 [Intermediate] • Target: Piecewise and absolute value review

Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.



11 [Intermediate] • Target: Piecewise and absolute value review

Solve the inequality $|3x + 1| \le 7$, writing your answer in interval notation.

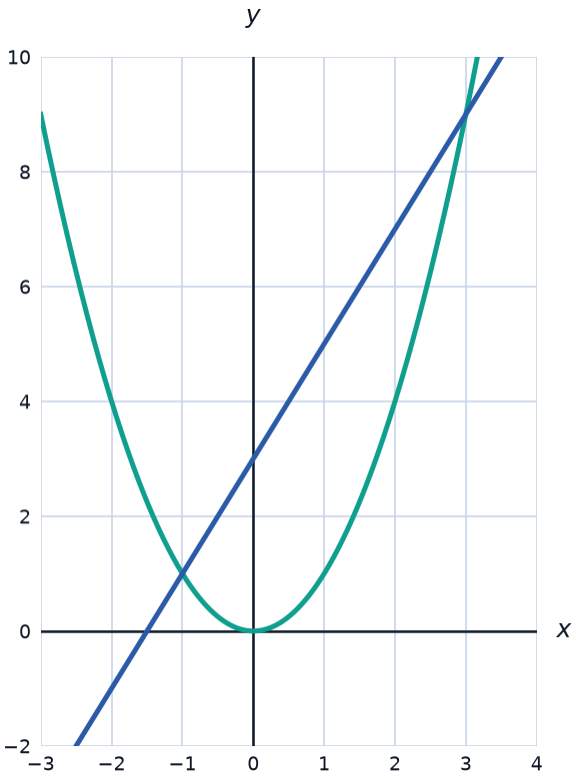
12 [Intermediate] • Target: Systems of equations

Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.

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13 [Intermediate] • Target: Systems of equations

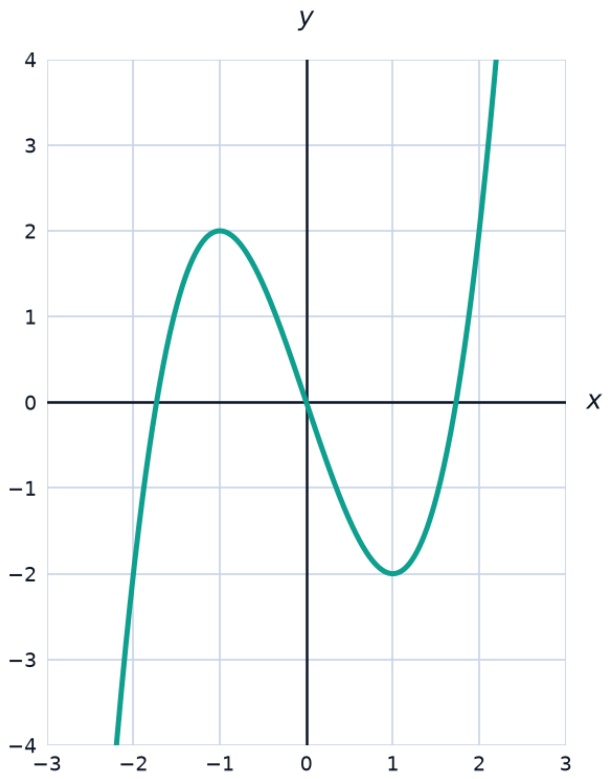
Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.



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14 **[Intermediate]** • **Target: Graphing functions calculus will build on**

The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the x-intercepts and the local max/min locations — features calculus will let you find exactly.

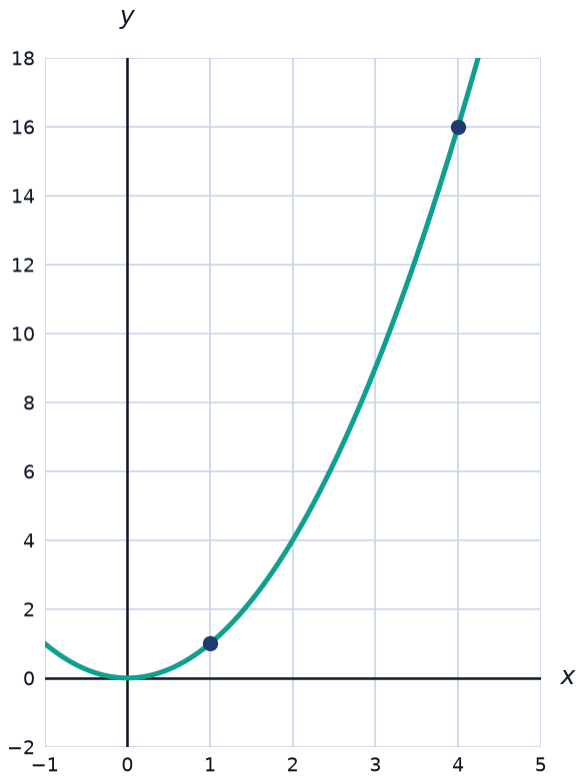


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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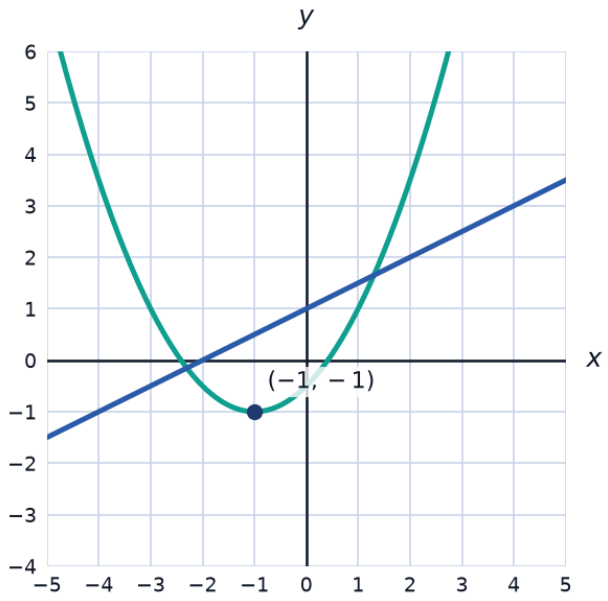
15 **[Challenging]** • **Target: Average rate of change (a preview of the derivative)** • **Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.



16 **[Challenging]** • **Target: Average rate of change (a preview of the derivative)** • **Cross-Lesson: 6.01 Related Rates**

For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of h, and describe what happens as $h \to 0$ (a preview of the derivative).



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17 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 6.02 Optimization and Curve Sketching

Simplify each expression, writing your answer with positive exponents:

a $x^{1/2} \cdot x^{1/3}$ **b** $(x^{2/3})^3$ **c** $\frac{x^{5/2}}{x^{1/2}}$

18 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 7.01 Integration by Substitution

Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.

19 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 7.02 Integration by Parts

Write the solution set of $x^2 - 4 > 0$ in interval notation.

20 [Challenging] • Target: Systems of equations • Cross-Lesson: 8.01 Area Between Curves

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Functions, Domain, and Composition

Given $f(x) = \sqrt{x - 3}$ and $g(x) = x^2 - 1$, find the domain of $(f \circ g)(x)$.

a. $(-\infty, -2] \cup [2, \infty)$ **b.** $[2, \infty)$

c. $[3, \infty)$ **d.** $[-2, 2]$

22 [Foundational] • Target: Functions, Domain, and Composition

If $f(x) = \frac{2x+1}{x-3}$, what is $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{x-2}$ **b.** $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$ **d.** $f^{-1}(x) = \frac{2x-1}{x+3}$

23 [Foundational] • Target: Functions, Domain, and Composition

Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = 3x^2 - 5x$.

a. $6x + 3h - 5$ **b.** $6x - 5$

c. $3x + 3h - 5$ **d.** $6x + h - 5$

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24 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Solve the equation $4^{x+1} = 8^{2x-1}$ for x.

a. 5/4

b. 1

c. 3/2

d. 2/3

25 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Solve $\ln(2x-1) + \ln(x) = \ln(3)$.

a. $x = 3/2$

b. $x = 1$

c. $x = 2$

d. $x = 3$

26 [Foundational] • Target: Exponentials, Logarithms, and Trig Review

Find the exact value of $\sin\left(\frac{7\pi}{6}\right) + \cos\left(\frac{4\pi}{3}\right)$.

a. -1

b. $-1/2$

c. 0

d. $-\sqrt{3}$

27 [Foundational] • Target: Rational Functions, Holes, and Asymptotes

Identify the vertical and horizontal asymptotes of $f(x) = \frac{3x^2 - 12}{x^2 - 5x + 6}$.

a. Vertical: $x = 3$; Horizontal: $y = 3$

b. Vertical: $x = 2, x = 3$; Horizontal: $y = 3$

c. Vertical: $x = 3$; Horizontal: $y = 0$

d. Vertical: $x = 2$; Horizontal: $y = 1$

28 [Foundational] • Target: Rational Functions, Holes, and Asymptotes

Determine the slant asymptote of $g(x) = \frac{2x^2 + 5x - 1}{x + 1}$.

a. $y = 2x + 3$

b. $y = 2x + 5$

c. $y = 2x - 1$

d. $y = x + 3$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Rational Functions, Holes, and Asymptotes**

Find the domain of $h(x) = \frac{\ln(x+2)}{\sqrt{9-x^2}}$.

a. $(-2, 3)$

b. $[-2, 3]$

c. $(-3, 3)$

d. $(0, 3)$

30 **[Intermediate]** • **Target: Trigonometric Identities and Equations**

Simplify $\frac{\sin(2\theta)}{1+\cos(2\theta)}$ into a single trigonometric function.

a. $\tan\theta$

b. $\cot\theta$

c. $\sec\theta$

d. $\sin\theta$

31 **[Intermediate]** • **Target: Trigonometric Identities and Equations**

Solve $2\cos^2 x + \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

a. $\pi/3, \pi, 5\pi/3$

b. $\pi/6, \pi, 11\pi/6$

c. $\pi/3, 2\pi/3$

d. $\pi/2, 3\pi/2$

32 **[Intermediate]** • **Target: Trigonometric Identities and Equations**

If $\sin\alpha = 3/5$ with $\alpha \in (\pi/2, \pi)$, find $\cos(2\alpha)$.

a. $7/25$

b. $-7/25$

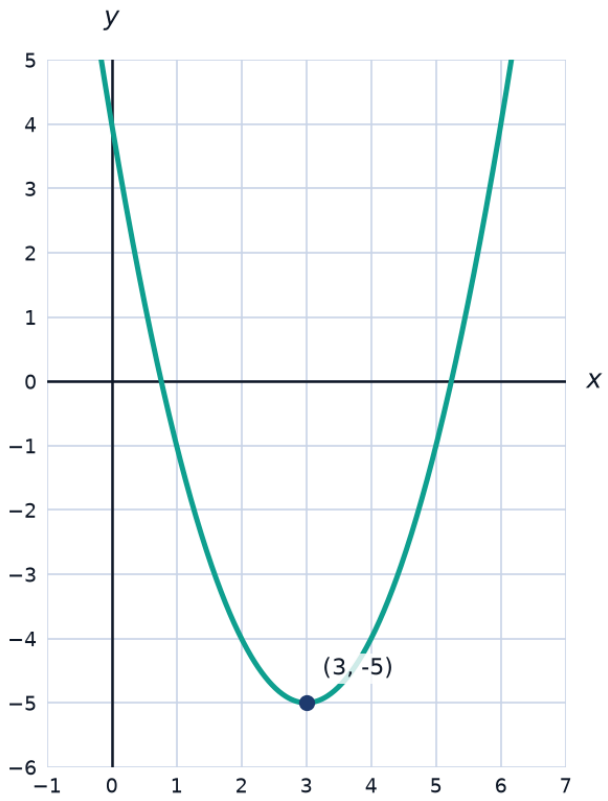
c. $24/25$

d. $-24/25$

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33 **[Intermediate]** • **Target: Rational Functions, Holes, and Asymptotes**

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.



a. -5

b. 4

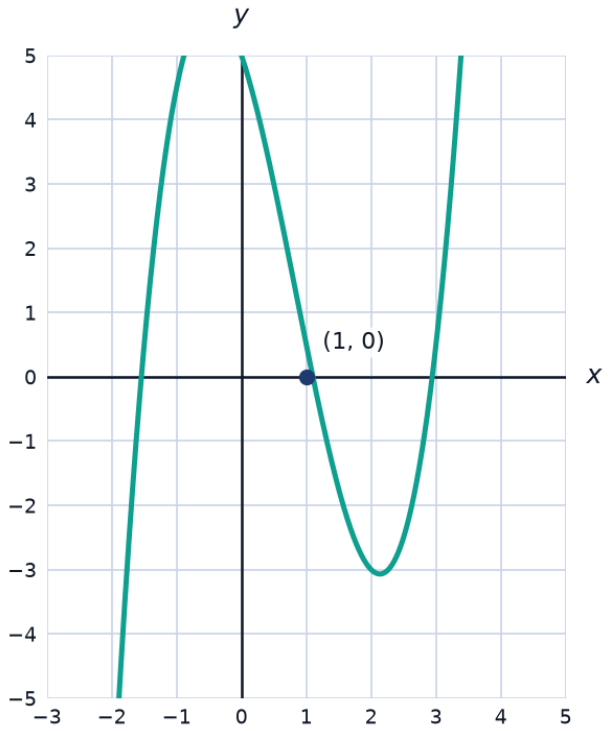
c. 9

d. 0

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34 [Intermediate] • Target: Trigonometric Identities and Equations

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?



a. $(-2, 3)$

b. $(0, -1)$

c. $(1, 0)$

d. $(3, 4)$

35 [Intermediate] • Target: Functions and composition

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

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36 [Intermediate] • Target: Lines, exponentials, and trig — the toolkit calculus builds on

What is the maximum rate of change of $f(x, y) = x\cos(y) + y$ at $(-2, -2)$?

a. $\sqrt{-4\sin(2) + 2 + 3\sin^2(2)}$

b. $-4\sin(2) + 2 + 3\sin^2(2)$

c. $-1 - \cos(2) + 2\sin(2)$

d. $-1 + 2\sin(2)$

37 [Intermediate] • Target: Factoring and rational expressions

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2, 3)$?

a. $3\sin^2(3) + 12\sin(3) + 10$

b. $2\sin(3) - \cos(3) + 3$

c. $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$

d. $2\sin(3) + 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Inverse functions and logarithms • Cross-Lesson: 7.01 Integration by Substitution

What is the maximum rate of change of $f(x, y) = x\cos(y) - 2y$ at $(3, -2)$?

a. $\sqrt{-12\sin(2) - 4\cos(4) + 9}$

b. $-12\sin(2) - 4\cos(4) + 9$

c. $-2 - \cos(2) + 3\sin(2)$

d. $-2 + 3\sin(2)$

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39 [Challenging] • Target: Piecewise and absolute value review • Cross-Lesson: 7.02 Integration by Parts

Consider the function $f(x) = 2x^2 - 3x + 1$. What is the value of $f(3)$?

a. 10

b. 13

c. 16

d. 19

40 [Challenging] • Target: Systems of equations • Cross-Lesson: 8.01 Area Between Curves

A spherical balloon is inflated at 4 cm³/s. Rate of radius change when $r = 2$:

a. $\frac{1}{\pi}$

b. $\frac{1}{2\pi}$

c. $\frac{1}{4\pi}$

d. $\frac{1}{8\pi}$

41 [Challenging] • Target: Graphing functions calculus will build on • Cross-Lesson: 8.02 Volumes of Revolution

What is the maximum rate of change of $f(x, y) = x^2 e^y + y$ at $(2, \ -1)$?

a. $\frac{16 + (e+4)^2}{e^2}$

b. $\frac{2 \cdot 2^{\frac{13}{476}} \cdot 3^{\frac{621}{952}} \cdot 5^{\frac{577}{952}} \cdot 7^{\frac{447}{952}}}{7}$

c. $\frac{\sqrt{16 + (e+4)^2}}{e}$

d. $1 + \frac{4}{e}$

42 [Challenging] • Target: Average rate of change (a preview of the derivative) • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Symmetric equations of line through (x_0, y_0, z_0) with direction (a, b, c):

a. $ax + by + cz = 0$

b. $x/a = y/b = z/c$

c. $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$

d. $x + y + z = 1$

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43 [Challenging] • Target: Radicals and rational exponents • Cross-Lesson: 9.01 Separable Differential Equations

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 5 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $500 - \frac{500}{e^{\frac{3}{20}}}$

b. 85

c. $\frac{10}{e^{\frac{3}{20}}} + 500$

d. $500 - \frac{490}{e^{\frac{3}{20}}}$

44 [Challenging] • Target: Interval notation and set operations review • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{10}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $100 - \frac{90}{e^{\frac{3}{10}}}$

b. $100 - \frac{100}{e^{\frac{3}{10}}}$

c. 40

d. $\frac{10}{e^{\frac{3}{10}}} + 100$

45 [Challenging] • Target: Functions, Domain, and Composition • Cross-Lesson: 11.01 Sequences and Series

A 1000-L tank holds brine with 10 kg of salt. Brine with $\frac{1}{5}$ kg/L enters at 25 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 30 min?

a. $200 - \frac{200}{e^{\frac{3}{4}}}$

b. 160

c. $\frac{10}{e^{\frac{3}{4}}} + 200$

d. $200 - \frac{190}{e^{\frac{3}{4}}}$

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